# MERN Stack Internship

Interview Questions — Fresher Level | 11 Topics | 110+ Questions

## JavaScript

01. What is the difference between var, let, and const?
02. What is hoisting in JavaScript?
03. Explain closures with an example.
04. What is the difference between == and ===?
05. What are Promises? How are they different from callbacks?
06. Explain async/await and how it works under the hood.
07. What is the difference between null and undefined?
08. What are arrow functions and how are they different from regular functions?
09. What is event delegation?
10. What is the difference between map(), filter(), and reduce()?
11. What is prototypal inheritance in JavaScript?
12. What is the 'this' keyword? How does its value change in different contexts?
13. What is a higher-order function? Give an example.
14. What is the difference between synchronous and asynchronous code?
15. What is destructuring? Give examples for arrays and objects.

## Node.js ★ Focus Area

■ This is one of your identified weak areas — study extra carefully.

01. What is the Node.js Event Loop? Explain each phase: timers, I/O, poll, check, close.
02. What is the difference between process.nextTick() and setImmediate()?
03. What are microtasks and macrotasks in the event loop?
04. Why is Node.js called single-threaded yet non-blocking?
05. What is the role of libuv in Node.js?
06. What is the difference between require() and ES6 import?
07. What are streams in Node.js? What are the 4 types?
08. What is the difference between blocking and non-blocking I/O?
09. What is the Node.js fs module? Difference between fs.readFile and fs.readFileSync?
10. What is the EventEmitter class and how does it work?
11. What is a buffer in Node.js and when would you use it?
12. What is middleware in the context of Node.js?
13. How do you handle errors in async Node.js code?

14. What is cluster module in Node.js and why would you use it?
15. What happens when you call setTimeout(fn, 0) — when does it execute?

## Express.js

01. What is Express.js and why do we use it over plain Node.js?
02. What is middleware in Express? Give an example.
03. What is the difference between app.use() and app.get()?
04. What are route parameters? How do you access them?
05. What is the difference between req.params, req.query, and req.body?
06. How do you handle errors globally in Express?
07. What is CORS and how do you enable it in Express?
08. What is the purpose of express.json() middleware?
09. How do you create a REST API with GET, POST, PUT, DELETE in Express?
10. What is the difference between res.send(), res.json(), and res.end()?

## React

01. What is the Virtual DOM and how does React use it?
02. What is the difference between a class component and a functional component?
03. What are React Hooks? Name the most common ones.
04. Explain useState with an example.
05. What does useEffect do? What are its dependency array cases?
06. What is prop drilling and how do you avoid it?
07. What is the Context API?
08. What is the difference between controlled and uncontrolled components?
09. What is React Router and how does it work?
10. What is reconciliation in React?
11. What are keys in React lists and why are they important?
12. What is the difference between useMemo and useCallback?

## MySQL

01. What is the difference between WHERE and HAVING?
02. What are the different types of JOINs? Explain with examples.
03. What is normalization? Explain 1NF, 2NF, 3NF.
04. What is the difference between DELETE, TRUNCATE, and DROP?
05. What is a primary key vs a foreign key?

06. What is an index and how does it improve performance?

07. What are ACID properties in a database?

08. What is the difference between UNION and UNION ALL?

09. What is a subquery? Give an example.

10. What is a stored procedure?

## MongoDB

01. What is the difference between SQL and NoSQL databases?

02. What is a document in MongoDB? How is it different from a table row?

03. What is the role of _id in MongoDB?

04. How do you perform CRUD operations in MongoDB?

05. What is the aggregation pipeline in MongoDB?

06. What is Mongoose and why is it used with Node.js?

07. What is the difference between embedding and referencing documents?

08. How do indexes work in MongoDB?

09. What is sharding in MongoDB?

10. What is the difference between find() and findOne()?

## Machine Learning ★ Focus Area

■ This is one of your identified weak areas — study extra carefully.

01. What is the difference between supervised, unsupervised, and reinforcement learning?

02. What is overfitting and underfitting? How do you prevent them?

03. What is the train-test split and why is it important?

04. What is a confusion matrix? Explain precision, recall, and F1-score.

05. What is the difference between classification and regression?

06. What is gradient descent? Explain in simple terms.

07. What is a feature? What is feature engineering?

08. What is cross-validation?

09. What is the difference between a parametric and non-parametric model?

10. What is a decision tree and how does it work?

## Deep Learning ★ Focus Area

■ This is one of your identified weak areas — study extra carefully.

01. What is a neural network? What are layers, neurons, and weights?

02. What is backpropagation?

03. What are activation functions? Name common ones (ReLU, Sigmoid, Tanh).
04. What is the vanishing gradient problem?
05. What is the difference between CNN and RNN?
06. What is dropout regularization?
07. What is batch normalization and why is it used?
08. What is transfer learning?
09. What is the difference between a loss function and an optimizer?
10. What frameworks are used for deep learning? (TensorFlow, PyTorch)

## Generative AI ★ Focus Area

■ This is one of your identified weak areas — study extra carefully.

01. What is a Large Language Model (LLM)?
02. What is the Transformer architecture? What is self-attention?
03. What is prompt engineering?
04. What is RAG (Retrieval-Augmented Generation)?
05. What is the difference between fine-tuning and prompt engineering?
06. What are tokens in the context of LLMs?
07. What is hallucination in AI models?
08. What is the difference between GPT, BERT, and T5?
09. What is an embedding in the context of NLP/GenAI?
10. What are vector databases and why are they used with LLMs?

## npm

01. What is npm and what is it used for?
02. What is the difference between dependencies and devDependencies?
03. What does npm install vs npm install --save-dev do?
04. What is package.json and what does it contain?
05. What is package-lock.json and why does it exist?
06. What is the difference between npm install and npm ci?
07. What is npx and how is it different from npm?
08. How do you define and run scripts in package.json?

## OOP (Object-Oriented Programming)

01. What are the 4 pillars of OOP? Explain each briefly.
02. What is the difference between a class and an object?

03. What is encapsulation and why is it important?
04. What is inheritance? Give an example.
05. What is polymorphism? Explain method overriding vs overloading.
06. What is abstraction?
07. How does JavaScript implement OOP? (prototype-based vs class-based)
08. What is the difference between public, private, and protected access modifiers?
09. What is a constructor?
10. What is the difference between composition and inheritance?

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Tip: For the Event Loop, know the execution order: process.nextTick → Promise.then (microtasks) → setTimeout/setImmediate (macrotasks). For ML/DL/GenAI, focus on concepts — interviewers rarely go deep at fresher level.